PRE-ALGEBRA (8TH) · UNIT 3: LINEAR AND FUNCTIONAL RELATIONSHIPS

Road Trip Rate Lab

Teacher Guide & Answer Keys

Teacher

2–3 class periods

Project Overview

Students compare three cars whose speeds are given by a table, a graph, and an equation, then plan a 500-mile trip with times and fuel costs.

The project builds fluency with slope as a rate, initial value, comparing representations, and recognizing nonlinear relationships.

Standards Addressed

- 8.EE.B.5 Graph proportional relationships; unit rate as slope
- 8.EE.B.6 $y = mx + b$
- 8.F.A.1–3 Define, compare, and identify linear functions
- 8.F.B.4–5 Construct and describe functions

Pacing (2–3 class periods)

Day	Plan
Day 1	Launch with the three cars. Part 1: rates from each representation.
Day 2	Part 2: comparing and catching up; gas function.
Day 3	Part 3: trip plan, cost, and comparison chart.

Materials

- Student handout with car cards
- Graph paper
- Calculators
- Optional: online map to choose a real destination

How the Three Versions Differ

Level	What changes
CP2	Rates from tables with starters, one equation per car, and a simple linear-or-not check.
CP1	Two-point slope, catch-up equations, interpreting slope in a fuel function, and total trip cost.
Honors	Functions from two points, delayed-start chases, a nonlinear stopping-distance model, and weighted average speed.

Watch For (Common Misconceptions)

- Dividing time by distance for a rate.
- Reading the y-intercept as the speed.
- Averaging speeds instead of dividing total distance by total time.

Extension Ideas

- Graph all cars on one set of axes and label the meeting points.
- Add a gas stop and model the tank as a piecewise function.
- Collect real speed data from a trip app.

Answer Key – CP2

Part 1: Car A from a Table

#	Task	Answer
1	What is Car A's rate of change (speed)?	55 miles per hour (distance increases 55 each hour).
2	Write an equation for Car A's distance d after t hours.	d = 55t
3	How far does Car A go in 4.5 hours?	55 × 4.5 = 247.5 miles.

Part 2: Comparing Cars

#	Task	Answer
4	Which car is faster, Car A or Car C? How do you know?	Car C: its rate 62 mph is greater than 55 mph.
5	How far from the start is Car B after 3 hours?	60(3) + 20 = 200 miles.
6	What do the 60 and the 20 mean in d = 60t + 20?	60 is the speed in miles per hour (slope); 20 is the starting distance (y-intercept): a 20-mile head start.

Part 3: Gas and Patterns

#	Task	Answer
7	Gas left after d miles is $g = 14 - \frac{d}{30}$. How much gas is left after 150 miles?	14 − 5 = 9 gallons.
8	How many miles can a car go before the tank is empty?	14 × 30 = 420 miles.
9	A square tile with side s has area A: s = 1, 2, 3 → A = 1, 4, 9. Is this relationship linear? Explain.	No. The area increases by 3, then 5; a linear relationship increases by the same amount each step.

Reflection

#	Prompt	Look for
R1	How can you find a rate of change from a table?	Divide the change in the output by the change in the input between two rows.
R2	What does the y-intercept mean on a distance graph?	It is the distance at time 0: where the car starts.

Answer Key – CP1

Part 1: Three Representations

#	Task	Answer
1	Write a function for Car A from the table. Is it proportional? Explain.	d = 55t. Yes: it passes through (0, 0) and the ratio d : t is always 55.
2	Car B's graph passes through (1, 80) and (3, 200). Find the slope and y-intercept, and write the function.	Slope = $\frac{200-80}{3-1}$ = 60. b = 80 − 60 = 20. d = 60t + 20.
3	Order the cars from fastest to slowest.	Car C (62 mph), Car B (60 mph), Car A (55 mph).

Part 2: Catching Up

#	Task	Answer
4	When does Car C catch Car B? How far have they driven?	62t = 60t + 20 → t = 10 hours; d = 620 miles.
5	The gas left is g(d) = 14 − $\frac{d}{30}$. What is the slope, and what does it mean?	Slope −$\frac{1}{30}$: the car uses $\frac{1}{30}$ gallon for each mile driven.
6	Is $f(t) = t^2 + 3$ a linear function? Use a table to explain.	No. f(0) = 3, f(1) = 4, f(2) = 7: the changes are 1, then 3, which are not constant.

Part 3: Plan the Trip

#	Task	Answer
7	Car A drives 55 mph and makes one 1.5-hour stop. How long is the trip, to the nearest tenth of an hour?	500 ÷ 55 ≈ 9.1 hours + 1.5 = 10.6 hours.
8	How much does gas cost for the 500-mile trip at 30 miles per gallon?	500 ÷ 30 ≈ 16.67 gallons × $3.60 = $60.00.
9	Write a function for gas cost c per mile driven d, and interpret its slope.	c = 0.12d ($3.60 ÷ 30 = $0.12 per mile). Each mile costs 12 cents.

Reflection

#	Prompt	Look for
R1	How can you compare two functions when one is a graph and one is an equation?	Find the rate of change and initial value of each; compare slopes for speed and intercepts for starting values.
R2	How can you tell from a table that a function is nonlinear?	Equal steps in the input give unequal changes in the output.

Answer Key – Honors

Part 1: Building Functions

#	Task	Answer
1	Car D was 130 miles from home after 2 hours and 310 miles after 5 hours. Write its function and interpret the initial value.	Rate = $\frac{310-130}{5-2}$ = 60 mph. d = 60t + 10: Car D started 10 miles from home.
2	After how many hours is Car D 50 miles ahead of Car A (d = 55t)?	(60t + 10) − 55t = 50 → 5t = 40 → t = 8 hours.
3	f(x) = 2x − 3 and g is given by the table (0, 1), (1, 4), (2, 7). Which has the greater rate of change? The greater initial value?	g: rate 3 > 2 and initial value 1 > −3. g is greater for both.

Part 2: Chases and Curves

#	Task	Answer
4	Car C (62 mph) leaves 1.5 hours after Car A (55 mph) from the same place. When does Car C catch up, and how far have they gone?	62(t − 1.5) = 55t → 7t = 93 → t = $\frac{93}{7}$ ≈ 13.3 hours after Car A left; d ≈ 730.7 miles.
5	Stopping distance is about d = 0.05v^2 feet at speed v mph. Find d at 20, 40, and 60 mph. Is the model linear?	20, 80, 180 feet. No: doubling speed from 20 to 40 multiplies distance by 4.
6	Car A costs $0.12 per mile; Car E gets 24 mpg with gas at $3.36. Which costs more for 500 miles, and by how much?	Car E: 3.36 ÷ 24 = $0.14 per mile → $70. Car A: $60. Car E costs $10 more.

Part 3: Average Speed

#	Task	Answer
7	A car drives 2 hours at 60 mph and 1 hour at 30 mph. What is its average speed?	(120 + 30) ÷ 3 = 50 mph.
8	Explain why the answer is not (60 + 30) ÷ 2 = 45 mph.	The car spends more time at 60 mph, so average speed must weight each speed by time: total distance ÷ total time.
9	Design a three-leg trip of 500 miles that arrives in exactly 9 hours. Give each leg's speed and time.	Answers vary. Example: 4 h at 60 mph (240), 3 h at 55 mph (165), 2 h at 47.5 mph (95): 500 miles in 9 hours.

Reflection

#	Prompt	Look for
R1	Why does a constant rate of change make a function linear?	Equal input steps add the same amount each time, so the points lie on a straight line.

#	Prompt	Look for
R2	When is average speed equal to the average of the speeds?	Only when the car spends equal time at each speed.